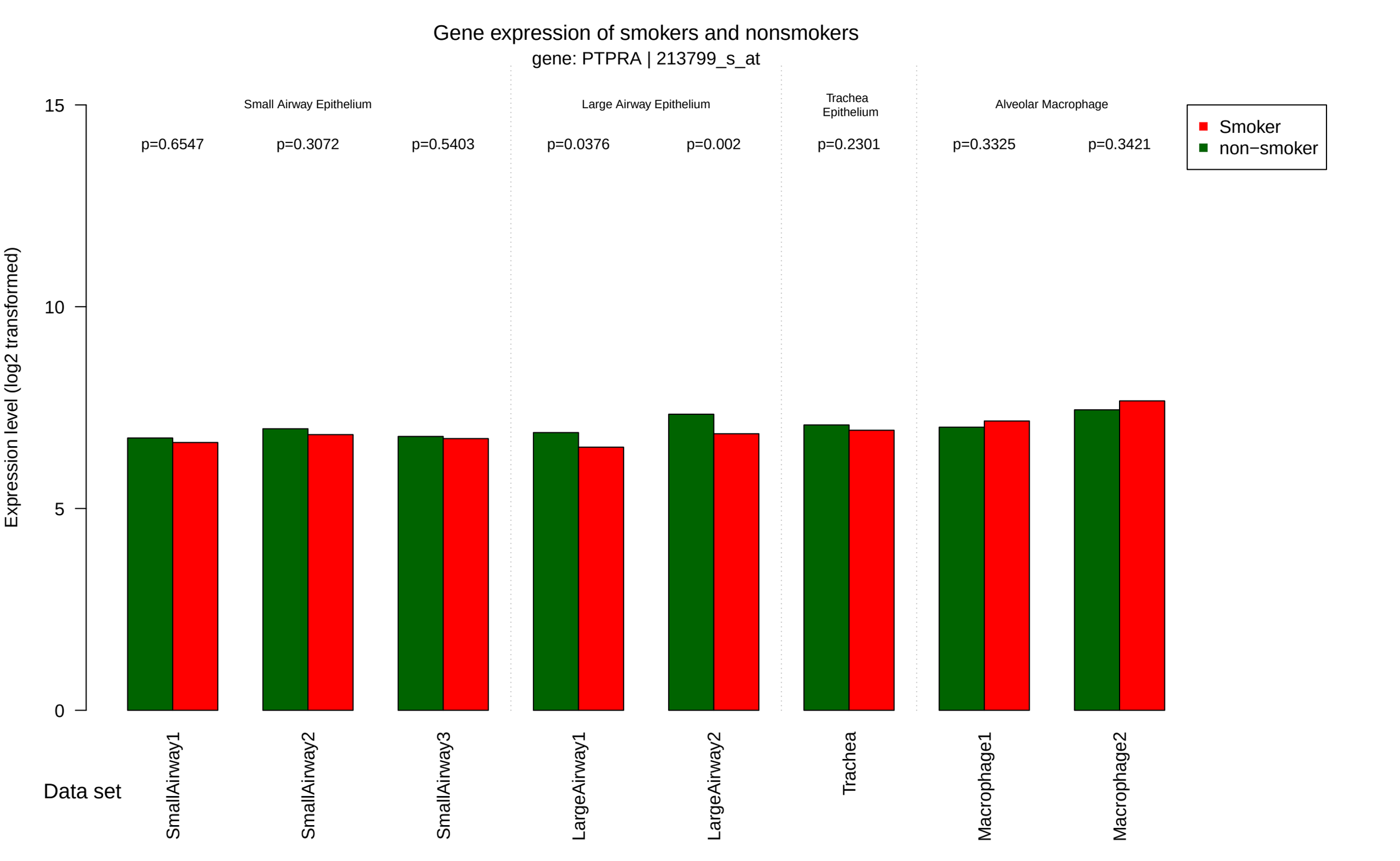
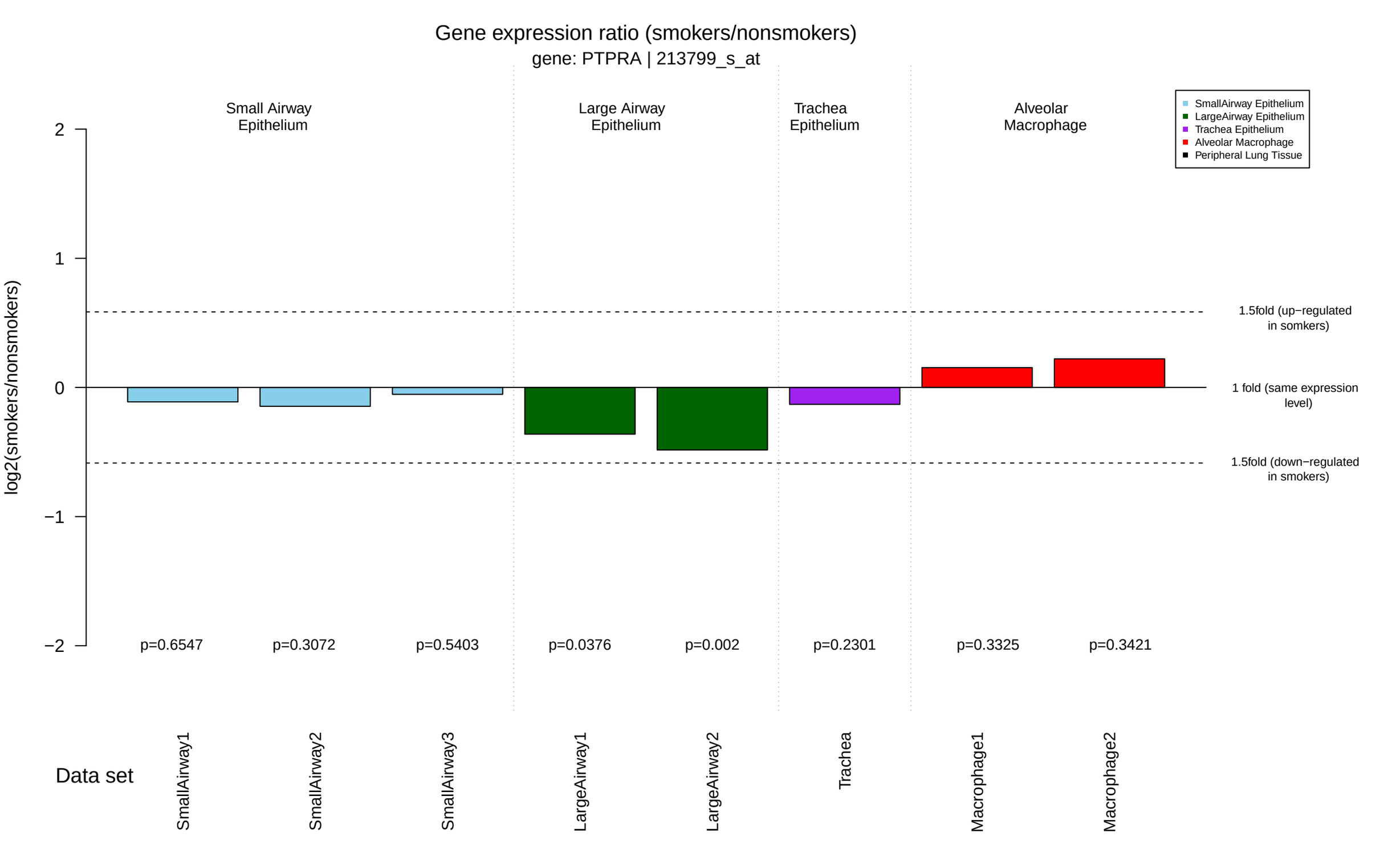
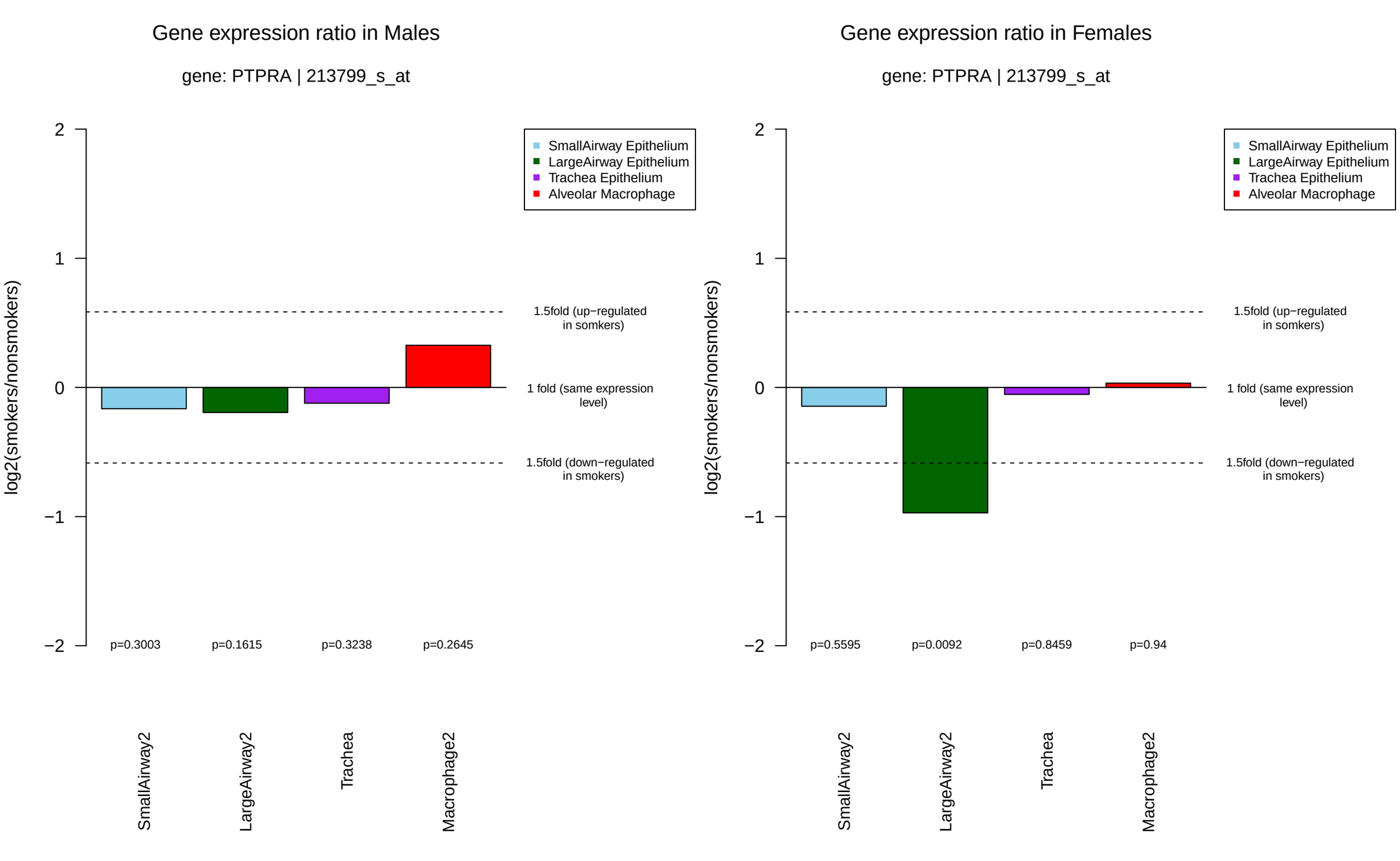
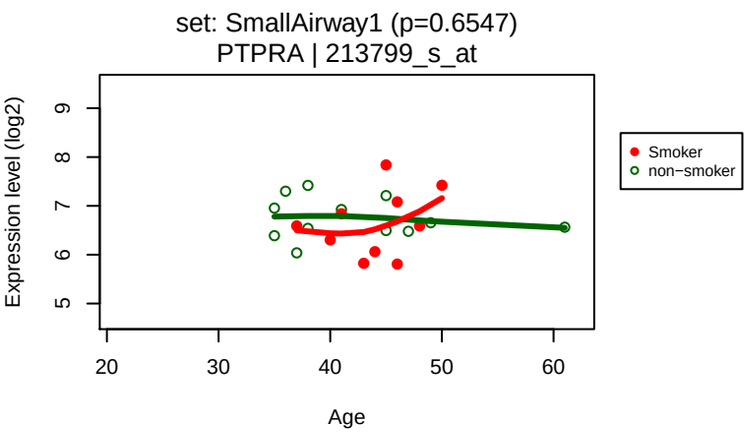
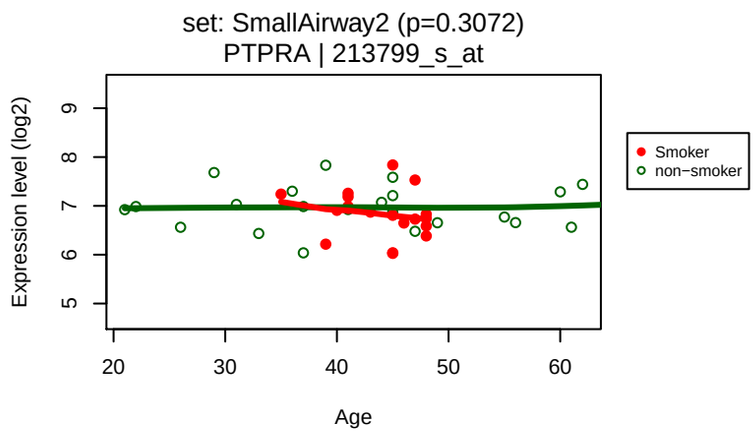
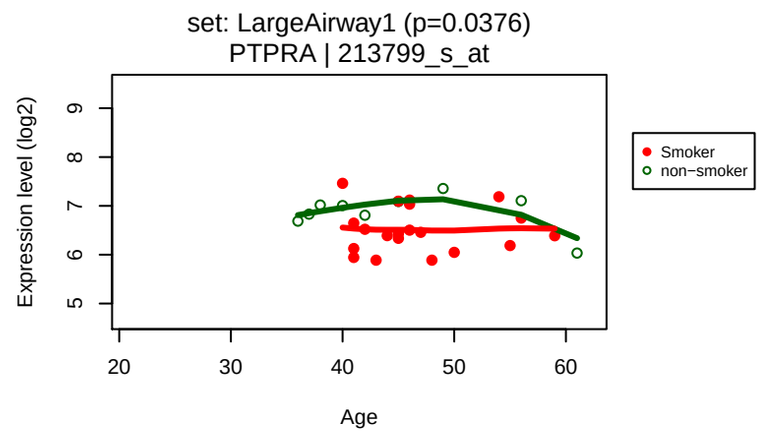

Gene expression of smokers and nonsmokers
gene: PTPRA | 213799_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6547
p=0.3072
p=0.5403
p=0.0376
p=0.002
p=0.2301
p=0.3325
p=0.3421
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

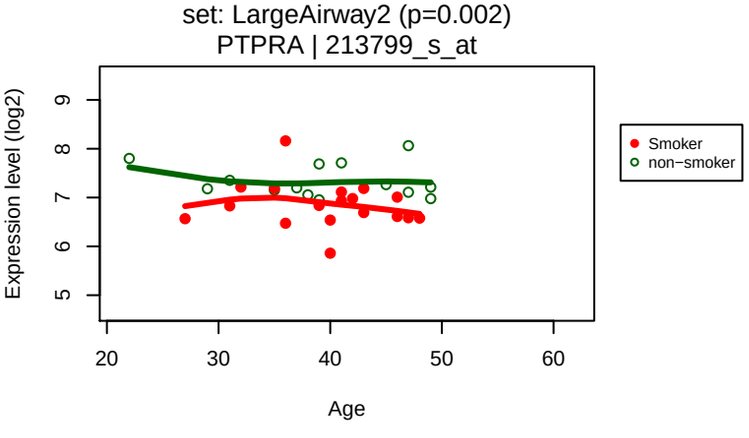

Gene expression ratio (smokers/nonsmokers)
gene: PTPRA | 213799_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6547
p=0.3072
p=0.5403
p=0.0376
p=0.002
p=0.2301
p=0.3325
p=0.3421
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

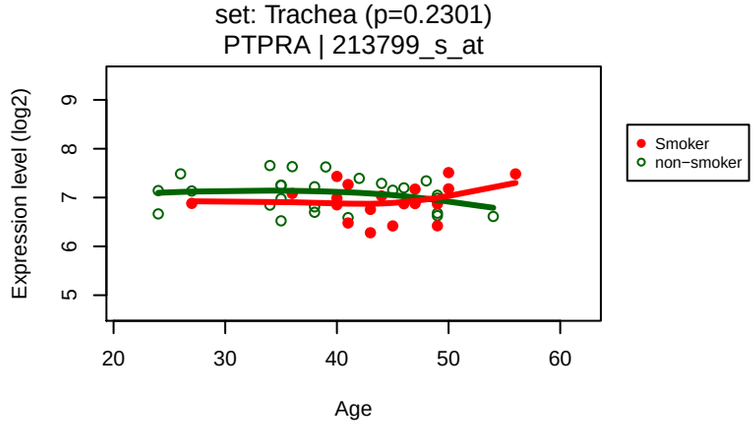

Gene expression ratio in Males
gene: PTPRA | 213799_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3003
p=0.1615
p=0.3238
p=0.2645
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPRA | 213799_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5595
p=0.0092
p=0.8459
p=0.94
SmallAirway2
LargeAirway2
Trachea
Macrophage2

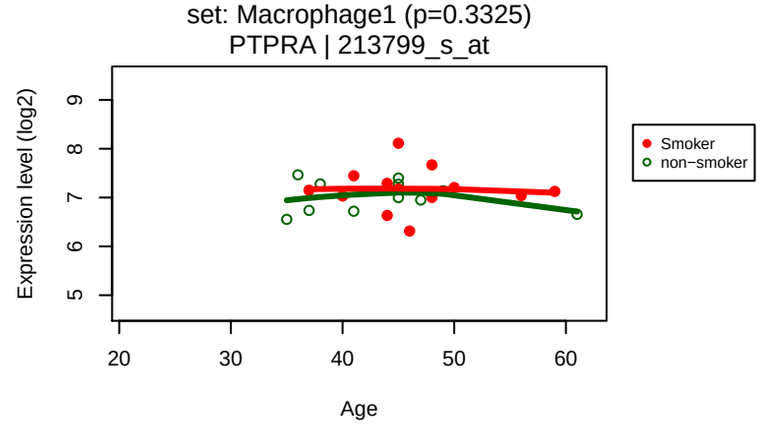

set: SmallAirway1 (p=0.6547)
PTPRA | 213799_s_at
Expression level (log2)
Age
Smoker
non−smoker

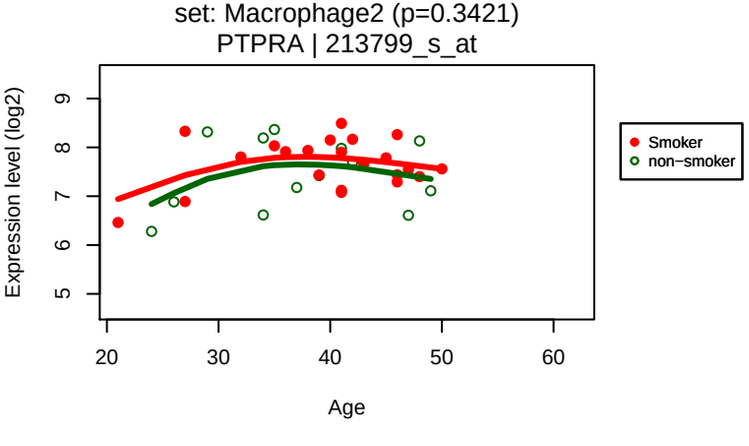

set: SmallAirway2 (p=0.3072)
PTPRA | 213799_s_at
Expression level (log2)
Age
Smoker
non−smoker

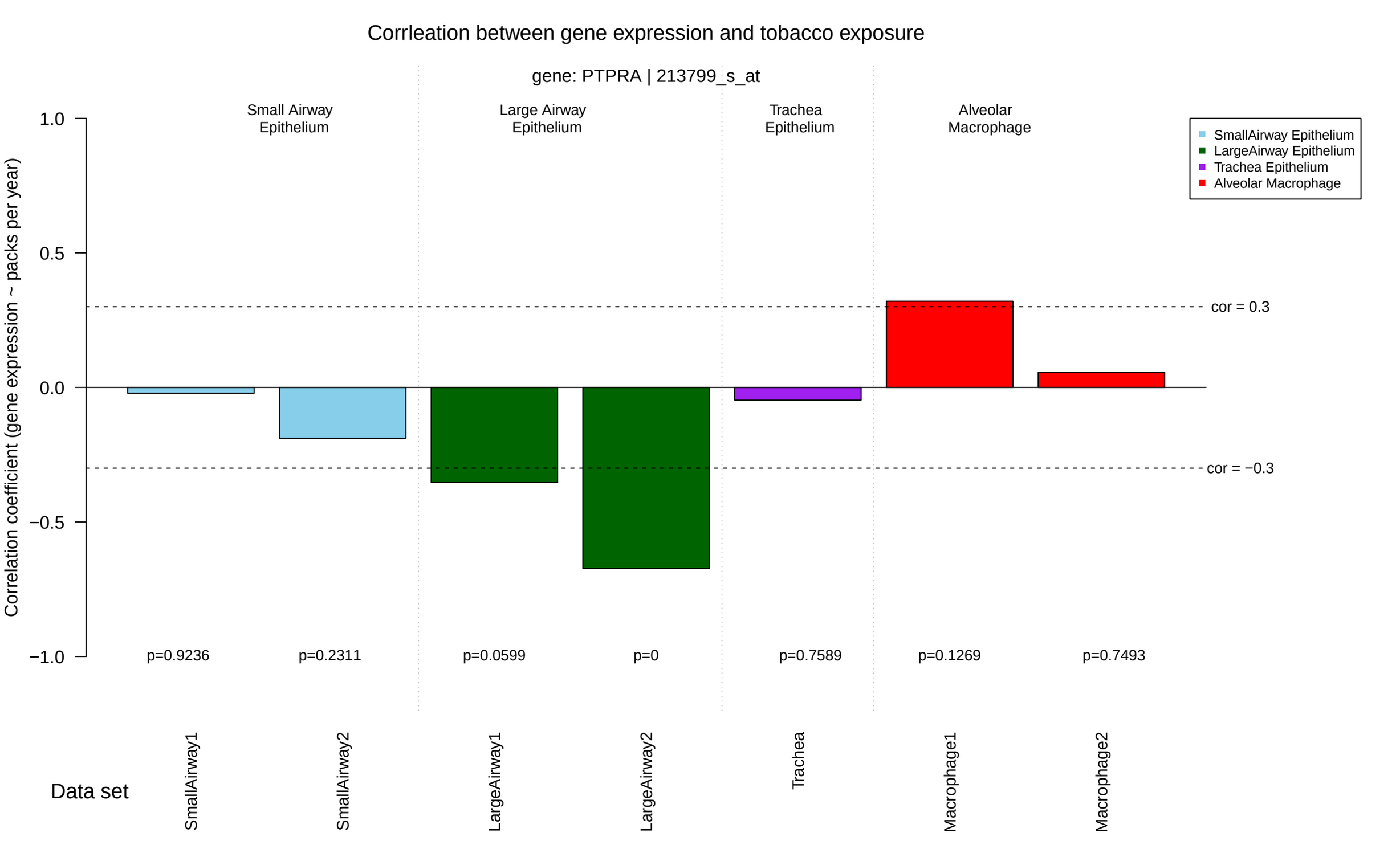

set: LargeAirway1 (p=0.0376)
PTPRA | 213799_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.002)
PTPRA | 213799_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2301)
PTPRA | 213799_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3325)
PTPRA | 213799_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3421)
PTPRA | 213799_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPRA | 213799_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9236
p=0.2311
p=0.0599
p=0
p=0.7589
p=0.1269
p=0.7493
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2